Case 2

Done	
Status	Cases

1. The cortico-hippocampal circuit

Core idea: Information from the entorhinal cortex (EC) reaches CA1, the main output of the hippocampus, along an indirect *trisynaptic* route and a *direct* route. Every excitatory stage is a site of plasticity.

- EC = main input; carries multimodal sensory and spatial information
- Indirect (trisynaptic) pathway — the information is re-coded at three successive excitatory synapses:
    1. EC layer II neurons → *perforant path* → excite dentate gyrus (DG) granule cells
    2. Granule cells → *mossy fibers* → excite CA3 pyramidal cells
    3. CA3 cells → *Schaffer collaterals* → excite proximal dendrites of CA1 pyramidal cells
    4. CA1 → back to deep EC layers and forward to the subiculum (links hippocampus to many brain regions)
- Direct pathway (temporoammonic): EC layer III → perforant path → distal apical dendrites of CA1 (and CA3), no intervening synapse
    - Direct and indirect inputs converge on CA1 → their interaction likely matters for storage/recall
- CA3 recurrent collaterals: CA3 cells strongly excite each other → basis of associative storage and recall (pattern completion, §12); in pathology → seizures
- CA2: input from EC layer II (direct + via DG/CA3) and from hypothalamic oxytocin/vasopressin neurons → output to CA1 = a third excitatory input to CA1

- Perforant path → DG synapse = where LTP was first described, and the synapse stimulated in the neurogenesis experiments (§21–22)

2. Long-term potentiation (LTP): definition and shared features

Core idea: Long-term memory is thought to be stored as lasting changes in the strength of connections within specific neuronal assemblies (unlike working memory, held by ongoing prefrontal activity). LTP — a persistent increase in EPSP amplitude after brief high-frequency (tetanic) stimulation — is the model for this change.

- Larger EPSP → higher probability that the postsynaptic cell fires
- Hebb's rule = the theoretical template: if cell A repeatedly helps fire cell B, A's efficiency in firing B grows ("cells that fire together wire together")
  - Key requirement = coincidence of pre- and postsynaptic activity
- LTP occurs at nearly all excitatory synapses of the circuit (§1)
- Persists for days–weeks in the intact animal, hours in slices
- LTP is a *family* of processes, not one mechanism
  - Shared by all forms:
    - *Homosynaptic*: induced by activity in the pathway that gets potentiated
    - *Synapse-specific*: only the stimulated synapses are potentiated
  - Differ in: receptors/channels required, second-messenger pathways, and whether expression is post-, pre- or both
- Two questions to keep apart:
  - *Induction* = the biochemical events the tetanus triggers (§3)
  - *Expression* = the lasting change that makes transmission stronger (§4)

3. NMDA-receptor-dependent LTP (Schaffer collateral → CA1): induction

Core idea: LTP at CA3→CA1 synapses is *associative* because the NMDA receptor conducts only when glutamate is bound **and** the postsynaptic membrane is depolarized. The Ca^{2+} that then enters the spine switches on kinases that strengthen the synapse.

NMDA receptor = the molecular implementation of Hebb's coincidence requirement:

1. CA3 terminal releases glutamate → binds AMPA and NMDA receptors on the CA1 spine
2. AMPA receptors (Na^+/K^+ only) open → depolarize the spine
3. At resting potential, extracellular Mg^{2+} plugs the NMDA pore → glutamate alone produces no NMDA current
4. Strong depolarization (tetanus, many coactive inputs) expels Mg^{2+} by electrostatic repulsion = **coincidence detector** (presynaptic glutamate + postsynaptic depolarization)
5. Unblocked NMDA channel conducts Ca^{2+} (plus Na^+, K^+) → large Ca^{2+} rise in the spine
6. Ca^{2+} activates CaMKII and PKC = **induction signal** → triggers the expression changes (§4)

- Size of the Ca^{2+} signal decides the outcome: large → kinases → LTP; small → phosphatases → LTD (§10)

Evidence: a tetanus given in the presence of the NMDA-receptor antagonist APV produces no LTP at Schaffer collateral synapses → induction here is fully NMDA-dependent.

4. Expression of LTP: more AMPA receptors (post) and more release (pre)

Core idea: At CA1 synapses, LTP is mostly expressed postsynaptically by inserting extra AMPA receptors, which can even turn "silent" synapses on. Stronger stimulation adds a presynaptic increase in glutamate release, signalled backwards from the postsynaptic cell.

- *Silent synapse* = has NMDA but no functional AMPA receptors

- At rest: Mg^{2+} blocks NMDA → no response → synapse looks absent
- Depolarized: NMDA current appears

Postsynaptic "AMPAfication" — turns a Ca^{2+} signal into a larger response to the same glutamate:

1. Strong stimulation releases glutamate at both nonsilent and silent synapses on one neuron
2. Nonsilent synapses: many AMPA receptors open → large depolarization spreads through the neuron
3. That depolarization expels Mg^{2+} from NMDA receptors at the silent synapses too (= associativity, §8)
4. Ca^{2+} enters the silent synapse's spine → PKC phosphorylates the tail of AMPA receptors stored in endosomal vesicles (reserve pool in the spine)
5. Vesicles fuse with the postsynaptic membrane → AMPA receptors inserted = **unsilencing** → synapse now transmits at rest

Presynaptic enhancement — needed because induction happens postsynaptically:

1. Postsynaptic Ca^{2+} / second messengers → postsynaptic cell releases a *retrograde messenger* (e.g. nitric oxide)
2. Messenger diffuses to the presynaptic terminal → activates presynaptic kinases
3. Kinases enhance glutamate release → larger EPSP
4. Only terminals that were active during the tetanus respond → synapse specificity preserved

Evidence (silent synapses): in paired recordings, a presynaptic spike gives no EPSC at rest but an NMDA EPSC when the postsynaptic cell is held depolarized; after pairing pre-spikes with postsynaptic depolarization, a fast AMPA EPSC appears at rest → LTP added AMPA receptors.

5. Other forms of LTP: mossy fiber and direct perforant path

Core idea: The same tetanus potentiates every pathway, but with different machinery. Mossy fiber LTP is presynaptic and needs no NMDA receptors or postsynaptic activity; direct perforant-path LTP uses NMDA receptors plus L-type Ca^{2+} channels.

Pathway	Induction depends on	Expressed	Associative?
Schaffer collateral → CA1	NMDA receptors	Mainly post (AMPA insertion)	Yes
Direct perforant path → CA1	NMDA receptors + L-type Ca^{2+} channels	Not described	Not described
Mossy fiber → CA3	Presynaptic Ca^{2+} → cAMP/PKA	Presynaptic (more release)	No

Mossy fiber LTP — strengthening without coincidence:

1. Tetanus → large Ca^{2+} influx into the mossy fiber terminal
2. Ca^{2+}/calmodulin activates adenylyl cyclase → cAMP rises
3. cAMP activates PKA
4. PKA phosphorylates presynaptic vesicle proteins → more glutamate released per spike → larger EPSP

- Postsynaptic activity not required → **nonassociative**, i.e. not Hebbian

Evidence: NMDA antagonist APV fully blocks Schaffer collateral LTP, only partly blocks direct-path LTP (APV + an L-type blocker blocks it fully), and leaves mossy fiber LTP intact; a PKA inhibitor blocks mossy fiber LTP.

6. Early vs late LTP

Core idea: A single tetanus gives *early LTP* (hours), carried by covalent modification of existing synapses. Repeated tetani give *late LTP* (a day or more), which needs gene transcription, new proteins and probably new synapses.

	Early LTP	Late LTP
Induced by	One train	Repeated trains
Lasts	Hours	≥ a day

	Early LTP	Late LTP
Based on	Phosphorylation of existing proteins	Transcription + protein synthesis; synapse growth
Protein-synthesis inhibitor	No effect	Blocked

Late LTP pathway — turns a transient Ca^{2+} signal into a lasting structural change:

1. Repeated tetani → prolonged Ca^{2+} influx into the postsynaptic cell
2. Ca^{2+} recruits adenylyl cyclase → cAMP → activates PKA
3. PKA activates MAP kinase → MAPK moves to the nucleus
4. MAPK phosphorylates CREB-1 = **the transcription switch**
5. pCREB-1 turns on CRE-containing genes → new mRNAs and proteins
6. New proteins → growth of new synaptic connections between co-active cells

- Early mechanisms differ between Schaffer and mossy fiber pathways, but late LTP uses this same cAMP–PKA–CREB route in both (as in long-term facilitation in *Aplysia*)
- pCREB also recruits CBP, a histone acetylase → lasting chromatin (epigenetic) changes in gene expression

Synaptic tagging and capture — explains how late LTP stays synapse-specific although proteins are made in the soma:

1. Strong stimulation of pathway A → new proteins made, available throughout the cell
2. Every activated synapse receives a local **tag** (e.g. a phosphorylation)
3. Only tagged synapses **capture** the new proteins → late LTP only there
4. Weak stimulation of pathway B sets a tag without protein synthesis → if it falls within the window when A's proteins are present, B also gets late LTP

- Consequence: events experienced close in time are stored in overlapping cell assemblies

Maintenance — what keeps late LTP going:

- **CaMKII = molecular switch**: brief Ca^{2+} → autophosphorylation (Thr286) → kinase stays active without Ca^{2+}
- **PKMζ** = atypical PKC isoform without a regulatory domain → constitutively active
    1. Tetanus increases translation of PKMζ mRNA already present in dendrites
    2. PKMζ stays active → keeps the extra AMPA receptors in the membrane
    - Another atypical PKC can substitute when PKMζ is deleted

Evidence: the protein-synthesis inhibitor anisomycin blocks late but not early LTP. A PKMζ inhibitor given during the tetanus blocks late LTP; given hours later it erases established late LTP → maintenance needs ongoing PKMζ activity.

7. Spike-timing-dependent plasticity (STDP)

Core idea: A more natural route to LTP than tetanus: pairing a single presynaptic input with a single postsynaptic spike at low frequency. The sign of the change depends on order — pre before post → LTP; post before pre → LTD.

- Explained by the NMDA receptor's coincidence detection (§3):
    - Pre → post: postsynaptic spike arrives during the EPSP, while glutamate is bound → Mg^{2+} expelled from an *open* receptor → large Ca^{2+} influx → LTP
    - Post → pre: depolarization comes before glutamate → Mg^{2+} relief happens while the channel is closed → small Ca^{2+} influx → LTD
    - Spikes far apart in time → no change
- Follows Hebb's rule, plus a corollary: active synapses that do *not* help fire the cell are weakened

8. Properties that make NMDA-dependent LTP a memory mechanism

Core idea: Because Mg^{2+} relief needs strong depolarization and glutamate binding is local, NMDA-dependent LTP is cooperative, associative and synapse-specific — each property matches a requirement of memory storage.

- **Cooperativity**: many inputs must be active together to depolarize enough to relieve the Mg^{2+} block
  - → only significant events (activating enough inputs) are stored; a weak input alone gives no LTP
- **Associativity**: a weak input paired with a strong input is potentiated, because the strong input's depolarization spreads to the weak synapse
  - → a meaningless event (CS) gains meaning when it coincides with a significant one (US), as in Pavlovian conditioning
  - → in recurrent CA3, links partly overlapping assemblies → related events become associated
- **Synapse specificity**: an inactive synapse binds no glutamate, so its NMDA receptors stay closed despite the cell's depolarization
  - → unrelated inputs are not strengthened
  - → far more information fits in a network when single synapses change than when a whole cell's excitability changes

9. Linking LTP to memory

Core idea: Blocking LTP impairs hippocampal memory, enhancing it improves memory, learning itself produces LTP-like changes, and wiping out the *pattern* of synaptic weights erases memory. Together: memory is stored as a specific pattern of potentiated synapses.

- Logic of the evidence:
  - Necessity: block LTP machinery → memory fails
  - Bidirectionality: boost LTP → memory improves
  - Occurrence: learning → LTP-like change that *occludes* later LTP (capacity is finite)
  - Specificity: uniform potentiation → memory lost (§14 builds on this)
- NMDA-receptor-dependent LTP is required for many hippocampal memories, not only spatial (object recognition, odor discrimination, social transmission of food preference)

Evidence:

- *Morris water maze* (the standard spatial paradigm): animal must find a hidden submerged platform using distal room cues → hippocampus-dependent; a visible/flagged platform version is hippocampus-independent (control); a *probe trial* (platform removed) measures time in the target quadrant.
- NMDA antagonist in hippocampus before training → hidden-platform learning fails, visible version intact; given after learning → recall intact → NMDA receptors needed for induction, not maintenance.
- NR1 (NMDA subunit) deleted only in adult CA1 → Schaffer collateral LTP abolished; slower learning and chance performance in the probe trial.
- NR2B overexpression → more Ca^{2+} through NMDA receptors → larger LTP and faster learning, better probe-trial memory.
- Reversible transgene (constitutively active CaMKII-Asp286, switched off by doxycycline): an intermediate-frequency tetanus gives LTD instead of LTP and spatial memory fails; switching the transgene off in the adult restores both → the deficit is not developmental.
- Learning induces LTP: after avoidance training, field EPSPs increase at a subset of CA1 sites, and later tetanus produces *less* LTP at exactly those sites (occlusion).
- Indiscriminate LTP across the DG after water-maze training erases the platform memory; blocking NMDA receptors before that stimulation prevents the loss.

10. Long-term depression (LTD)

Core idea: If synapses could only strengthen, the network would saturate and memories would blur together. LTD — a lasting weakening induced by prolonged low-frequency stimulation (or post-before-pre timing) — counteracts LTP, and in its main form uses the same NMDA receptors.

NMDA-dependent LTD — same receptor as LTP, opposite outcome, decided by how much Ca^{2+} enters:

1. Prolonged low-frequency stimulation → only modest postsynaptic depolarization
2. Mg^{2+} block only weakly relieved → **small** Ca^{2+} rise through NMDA receptors
3. Small Ca^{2+} is too little for CaMKII but enough for the high-affinity phosphatase **calcineurin** (and PP1)
    - Glutamate binding may also act *metabotropically* on the NMDA receptor → directly activates PP1
4. Phosphatases dephosphorylate synaptic proteins → **AMPA receptor endocytosis**
5. Fewer AMPA receptors → smaller EPSP

- mGluR-dependent LTD = distinct form: G-protein-coupled metabotropic glutamate receptors → MAP kinase (not phosphatases) → less presynaptic release + altered AMPA trafficking
- Functions: prevents LTP saturation; keeps memories specific; allows flexible updating (and reversal of learned fear in the amygdala)

Evidence: an NMDA antagonist blocks low-frequency LTD. Mice expressing a phosphatase (PP2A) inhibitor lack NMDA-dependent LTD but keep LTP; they learn a water-maze platform normally, but when it is moved they keep searching at the old location → LTD is needed for flexibility, not initial learning.

11. Memory is stored in cell assemblies (engrams)

Core idea: A memory is held by a *cell assembly* — the neurons active during encoding, bound together by synapses strengthened (via LTP) at that time. Reactivating that assembly is both sufficient and necessary for recall.

- *Engram* = the lasting physical trace of a memory
- Different memories = different assemblies → only a small subset of synapses changes per memory (why indiscriminate LTP erases memory, §9)

Engram tagging — lets an experimenter label and later control only the cells active during one experience:

1. Experience activates a subset of DG granule cells → Ca^{2+} → CaMK → pCREB → *c-fos* expression (activity marker)
2. In the transgenic mouse, c-Fos drives expression of the transcription factor tTA
3. tTA switches on a virally delivered TRE-ChR2 gene (light-activated cation channel) — but only when doxycycline is absent (doxycycline binds and inhibits tTA) = **the time window**
4. Doxycycline removed during the experience → only cells active then express ChR2, which persists for days
5. Doxycycline restored → no further labeling; blue light can now reactivate just that assembly

Evidence:

- Sufficiency: mice fear-conditioned in context B with tagging on; light delivered to DG in a neutral context A → freezing (no freezing with light off, or in unconditioned mice).
- Necessity: silencing (light-activated Cl^- pump) the CA1 cells active during conditioning blocks freezing in the conditioning context.
- False memory: cells active in a safe context A are tagged; they are light-reactivated while the mouse is shocked in context B → the mouse later freezes in A, where it was never shocked → an assembly's meaning can be rewritten by pairing it with a new experience.

12. Division of labour: DG separates, CA3 completes, CA2 recognizes others

Core idea: The hippocampal subregions solve opposite problems. The DG makes similar inputs *more different* so memories don't interfere (pattern separation); CA3 restores a whole stored pattern from a fragment (pattern completion); CA2 stores social memory.

Dentate gyrus — pattern separation

- Problem: two similar experiences (e.g. similar contexts) must be stored as distinct patterns

- Mechanism = *expansion recoding*: a limited number of EC inputs diverge onto a much larger population of granule cells → overlapping input patterns become less overlapping output patterns
- Activity patterns for two environments differ more in DG/CA3 than one synapse upstream in EC
- DG lesions or DG-specific manipulations → animals can't discriminate similar locations/contexts
- Adult-born granule cells appear especially important for this → §23

CA3 — pattern completion (recall of a complex memory from a few cues):

1. Encoding: co-active CA3 cells strengthen their recurrent synapses (associative LTP) → a stored assembly
2. Retrieval: a partial cue reactivates only part of the assembly
3. Strong recurrent connections spread activity from that part to the rest
4. The whole original pattern is reinstated = **pattern completion**

Evidence: deleting NMDA receptors only in CA3 abolishes LTP at the recurrent synapses alone; these mice find the water-maze platform normally with all spatial cues but fail when most cues are removed.

CA2 — social memory

- Genetic silencing of CA2 → mouse no longer remembers other mice; object and place memory intact
- CA2 has very high oxytocin and vasopressin receptor levels; stimulating its vasopressin input prolongs social memory
- Output → ventral CA1 (emotion-related), also needed for social memory

13. The spatial map: place cells and entorhinal cells

Core idea: Hippocampal *place cells* fire at specific locations and together form an allocentric *cognitive map* that is rebuilt (remapped) for each environment. The medial entorhinal cortex supplies a rigid, universal spatial metric through grid, head-direction, border and speed cells.

Place cells (CA1, CA3)

- Each fires when the animal is in its *place field*; different cells → different fields → the active set reads out position
- Allocentric: fixed to the outside world, not to the body (unlike sensory maps)
- New environment → new fields within minutes, stable for weeks
- *Remapping*: a major (sometimes minor) change of environment → fields shift to unrelated locations → each environment gets its own uncorrelated map
- *Replay*: during sleep after running a track, place cells fire again in the same order → stored trace of past trajectories

Medial entorhinal cell types (the input side)

- **Grid cells**: fire at many locations forming a hexagonal lattice → population always signals current position, in any environment, even in darkness
  - Neighbouring cells differ in grid phase (x,y offset)
  - Grid spacing grows from dorsal to ventral EC in discrete steps → modular organization
- **Head-direction cells**: fire when facing one direction; some are conjunctive grid × direction cells
- **Border cells**: fire near walls/edges → anchor the grid to the enclosure's geometry (object-vector cells do the same relative to landmarks)
- **Speed cells**: firing proportional to running speed
- Direction + speed = velocity → lets the grid update position from self-motion

	Medial EC (grid etc.)	Hippocampus (place cells)
Across environments	Rigid: cell-pair relations preserved	Remaps: maps uncorrelated
Good for	Fast, accurate position (universal metric)	Storing many discrete memories with little *interference*

- Dorsoventral gradient: grid spacing and place-field size both increase dorsal → ventral (dorsal EC projects to dorsal hippocampus); dorsal hippocampus = spatial memory, ventral = nonspatial, social and emotional memory

Stability of place fields needs late LTP and attention

- LTP is not needed to *form* place fields but to *stabilize* them
- Attention stabilizes the map via the late-LTP pathway (§6):
    1. Animal attends to space (e.g. must run to a goal location)
    2. Dopamine acts on D1/D5 receptors in the hippocampus
    3. D1/D5 → cAMP → PKA
    4. PKA promotes late LTP → place fields stable for days
    - Without attention: fields form but drift within hours, and the animal fails spatial tasks

Evidence: without NMDA receptors, place fields still form but are larger and fuzzier; with late LTP blocked (PKA inhibitor), fields form but are stable only for about an hour.

14. The saturation problem and its homeostatic solutions

Core idea: Every hippocampus-dependent learning event strengthens synapses via an LTP-like mechanism, so a lifetime of everyday learning would drive overall synaptic weight toward saturation and leave no room for new memories. The hippocampus must therefore keep renewing its *memory capacity* — the ability to learn new events — through mechanisms that scale synaptic strength back down.

- *Memory capacity* = ability to acquire/form new memories (learning capacity), not retention of old ones
- Why saturation blocks learning:
    - Synapses already at maximum → no further LTP can be induced → new information cannot be written
    - Uniformly strengthened network → stored *pattern* of strong vs weak synapses is lost (§9)
- Three counter-mechanisms, on different scales:
    - **LTD** — weakens individual synapses, balancing LTP locally (§10)

    - **Sleep** — removes and shrinks spines → scales excitatory synapses down = **fast** downscaling
    - **Adult neurogenesis** — gradually renormalizes strong, long-lasting LTP back to baseline = **slow** downscaling (§22)
- Sleep and neurogenesis are complementary: neurogenesis clears the strong, consolidated potentiation that escapes sleep-mediated erasure

Evidence (LTP saturation impairs learning): the perforant path → DG pathway of rats is stimulated with repeated high-frequency trains until the fEPSP plateaus (or whole-brain repeated electroconvulsive shocks are given). One day later, rats learn *contextual* fear conditioning (shock in a chamber; freezing on re-exposure to it = hippocampus-dependent) poorly, but *auditory* fear conditioning (tone + shock; freezing to the tone in a new chamber = hippocampus-independent control) normally → saturation removes specifically hippocampal learning capacity, not general fear or activity.

15. Neural progenitors: self-renewal and modes of division

Core idea: Neurons and glia come from progenitor (stem) cells that both copy themselves (*self-renewal*) and produce differentiated daughters. The balance between symmetric and asymmetric division, set by local signals, decides how large the progenitor pool gets and how many cells it produces.

- Stem-cell properties = self-renewal + multipotency (neurons, astrocytes, oligodendrocytes)
- Proliferative zones around the ventricles (ventricular zone) = main source of CNS cells
- Two modes of division:
    - **Symmetric** → two progenitors = expands the pool
    - **Asymmetric** → one progenitor + one differentiated daughter = produces cells while the pool stays constant
- Local signals control the outcome in two ways (a distinction used again in §18–21, §27):

- **Instructive**: bias the division → one fate at the expense of others
- **Selective**: allow only certain progeny to survive and mature

Radial glial cells = progenitors and scaffold

- Cell body in the ventricular zone; one long process anchored at the pial surface
- Divide asymmetrically and self-renewingly → major source of neurons and astrocytes
- The long process = guide rail for neurons migrating out of the ventricular zone (§17)
- After neurogenesis ends, most become astrocytes
- May also be the progenitors of neurons in the adult brain → adult stem cells have glial (astrocytic) character (§20)

Evidence: when single radial glial cells are labeled with dyes or viruses, their clones contain both neurons and radial glia → radial glia are self-renewing progenitors, not only a scaffold.

16. Neuron or glia: Delta–Notch and bHLH transcription factors

Core idea: Neighbouring progenitors start out equivalent. Delta–Notch signalling amplifies a tiny difference between them into an all-or-none decision: the cell with **low** Notch activity becomes a neuron, the cell with **high** Notch activity is diverted to a progenitor or glial fate. Proneural bHLH transcription factors carry out the neuronal program.

Lateral inhibition — turns a small random imbalance into a clean fate split:

1. Neighbouring proneural cells express Delta (ligand) and Notch (receptor) at similar levels → all could become neurons
2. One cell presents slightly more Delta → Notch in its neighbour is activated more strongly
3. Delta binding → Notch's cytoplasmic domain is cleaved (Notch-intra) → moves to the nucleus = **messenger to the genome**

4. Notch-intra (with suppressor of hairless) activates the bHLH repressor *enhancer of split*
5. Enhancer of split represses the proneural bHLH gene *achaete-scute* → the neighbour makes less Delta
6. The first cell now receives less Notch signal → more achaete-scute → more Delta → the loop feeds itself = **amplifier**
7. Outcome: low-Notch cell → neuron; high-Notch cell → non-neuronal fate

What high Notch means depends on the stage (mammalian cortex):

- Early: Notch → Hes1/Hes5 (bHLH repressors) → keep radial glial identity (via an ErbB receptor for neuregulin)
    - Delta1 and neuregulin are made by newborn neurons → the progenitor pool is maintained by feedback from its own progeny
- Late: Notch → Hes → STAT3 (+ JAK2) → astrocyte differentiation (GFAP)
- Oligodendrocytes: Notch regulates the bHLH factors Olig1/Olig2; in glial progenitors Notch favours astrocytes over oligodendrocytes

Becoming a neuron = escaping Notch:

1. Numb (cytoplasmic Notch antagonist) is segregated into the neuronal daughter of an asymmetric division
2. Notch blocked → proneural bHLH factors expressed (Mash1, neurogenin-1/2)
3. Neurogenins activate downstream bHLH factors (NeuroD) → neuronal differentiation
4. Neurogenins also inhibit JAK–STAT → astrocyte fate blocked

- Loss of numb → progenitors keep proliferating
- Brakes that must also be removed: REST/NRSF (represses neuronal genes in progenitors and glia; degraded as neurons differentiate) and SoxB factors (keep progenitors by blocking neurogenic bHLH)
- Link: in adult-born granule cells this decision is made early (Prox-1/NeuN) and LTP does not change it (§20–21)

17. Migration: getting the new neuron to its place

Core idea: Newborn neurons leave the proliferative zone and travel to their final position, which determines the circuit they join. Central neurons migrate along scaffolds: radial glia (*radial migration*) or axon tracts and other neurons (*tangential* and *chain migration*). Peripheral neural crest cells migrate without a scaffold (*free migration*).

Radial migration (cortex)

Nucleus–centrosome coupling — how a neuron climbs a radial glial fiber:

1. The leading process wraps the radial glial fiber and extends along it (adhesion via integrins)
2. A microtubule cage forms around the nucleus; the centrosome moves forward into the leading process
3. The dynein motor complex (Lis1, Ndel1), on microtubules stabilized by doublecortin, pulls the nucleus toward the centrosome = **the motor**
4. The trailing process retracts → the cycle repeats → stepwise movement

- Leading process lacks the actin structure of a growth cone → closer to an extending dendrite than to an axon
- **Inside-first, outside-last**: neurons that leave the cell cycle early settle in deep layers; later-born ones migrate past them to superficial layers → a neuron's layer is set by its *birthday*
- **Reelin** (secreted by Cajal–Retzius cells) = stop signal that lets neurons detach from radial glia
  - No reelin → neurons pile up under the cortical plate → layering inverted
- Lis1 or doublecortin mutations → migration fails → lissencephaly (fewer, disordered layers)

Tangential and chain migration

- Neurons use axon tracts as guides → adds neuron types from distant birthplaces
- Cortex therefore has two sources:

    - Excitatory pyramidal neurons ← cortical ventricular zone (radial)
    - Inhibitory interneurons ← medial ganglionic eminence (tangential)
- Lateral ganglionic eminence → olfactory bulb interneurons via the *rostral migratory stream*
    - *Chain migration*: neurons use each other as substrate
    - In the adult the same stream is fed from the subventricular zone (§20)

Free migration (neural crest → PNS)

- BMP signals → snail/slug/twist → epithelial-to-mesenchymal transition; cadherins down, integrins up → cells delaminate from the neural tube
- Ephrin-B in the posterior half of each somite repels crest cells → channels them through the anterior half

18. Transmitter identity: built-in code vs target-induced switch

Core idea: In central neurons the transmitter (glutamate vs GABA) is fixed by bHLH transcription factors already expressed in the progenitors, i.e. decided before the neuron is born. In peripheral (sympathetic) neurons the transmitter stays plastic and can be switched by signals from the target the axon reaches.

Central neurons: a bHLH code tied to birthplace

Region	Excitatory (glutamatergic)	Inhibitory (GABAergic)
Cerebral cortex	Neurogenin-1/2; cortical ventricular zone	Mash1 (+ Dlx1/2); ganglionic eminences
Cerebellum	Math-1; rhombic lip (granule cells)	Ptf1a; ventricular zone (Purkinje cells etc.)

- Factors are in neuroepithelial cells, not mature neurons → identity specified before neuronal birth
- Neural crest uses the same logic: neurogenins → sensory neurons; Mash1 → sympathetic neurons

Sympathetic neurons: the target can switch the transmitter

Target-induced switch — matches the transmitter to what the target needs:

1. BMPs from the dorsal aorta induce Mash1, Phox2 and Gata2 → neurons start noradrenergic
2. Axons reaching sweat glands are exposed to gp130 (IL-6-class) cytokines: CNTF, LIF, cardiotrophin-1 = **instructive target signal**
3. Neurons stop making norepinephrine and dense-core vesicles
4. They start making acetylcholine and small clear vesicles → cholinergic

Evidence: sweat glands transplanted into a region normally innervated by noradrenergic fibers make those neurons cholinergic → the target supplies the switching signal.

19. Survival: neurotrophic factors hold back a latent death program

Core idea: Neurons are overproduced, and about half die during normal development. A neuron survives only if it receives enough of a neurotrophic factor, which its target supplies in limited amounts. The factor does not feed the cell; it suppresses an apoptotic program that every cell carries.

- **Neurotrophic factor hypothesis**: target cells secrete small amounts of a trophic factor → nerve terminals take it up → neurons that get too little die
  - → neuron number adjusts to target size
- Neurotrophins: NGF, BDNF, NT-3 (other families too: GDNF, IL-6-like cytokines, TGF-β, FGFs)
- Signal travels from terminal to soma: neurotrophin–Trk complex internalized → carried retrogradely in *signaling endosomes*

Two receptor classes

	Trk receptors	p75
Binds	One neurotrophin each: TrkA–NGF, TrkB–BDNF, TrkC–NT-3	All neurotrophins
Type	Receptor tyrosine kinase (dimer)	TNF-receptor family

	Trk receptors	p75
Effect	Survival + differentiation	With neurotrophin: survival (NF-κB); without: death (caspases)

- Trk signaling splits into two outputs:
  - Ras → MAPK → **differentiation**
  - PI3K → Akt → **survival**

Intrinsic apoptosis — the default that trophic signals keep switched off:

1. Loss of trophic support (or DNA damage, anoxia) shifts the Bcl-2 family balance toward pro-apoptotic members (BAX, BAK) = **the decision point**
2. BAX/BAK permeabilize the mitochondrial outer membrane → cytochrome c released into the cytosol
3. Cytochrome c binds Apaf-1 → activates the initiator caspase-9
4. Caspase-9 cleaves and activates effector caspases (caspase-3, -7) = **executioners**
5. Effector caspases cleave many proteins → DNA fragmentation, chromatin condensation, blebbing, phagocytosis → apoptosis

- Trophic brake: Trk → PI3K → Akt → promotes Bcl-2-like (anti-apoptotic) activity and inhibits caspase-9 → step 1 never tips
- Extrinsic route: death receptors (Fas, TNF receptor) → caspase-8 → the same effector caspases
- Apoptosis (active program, cell shrinks and is removed) ≠ necrosis (acute injury, rapid membrane lysis, no program)
- Conserved from worms: ced-9 = Bcl-2, ced-4 = Apaf-1, ced-3 = caspase
- Links: the survival window of adult-born granule cells is this same selection, and LTP-induced BDNF is a candidate trophic signal (§21); axotomy cuts neurons off this supply (§24)

Evidence:

- Removing a limb bud in the chick embryo leaves far fewer motor and sensory neurons; grafting an extra limb leaves more → the target controls how many

neurons *survive*, not how many are produced.

- Protein-synthesis inhibitors prevent the death of sympathetic neurons deprived of NGF → death is an active program that the trophic factor suppresses.

20. Adult neurogenesis: development rerun in two niches

Core idea: In two niches of the adult brain, stem cells keep producing neurons that run through the developmental program of §15–19: proliferate, commit to a neuronal fate, migrate, integrate into circuits, and survive only if selected. New neurons gradually replace older ones, so the total number in these regions stays roughly constant.

- Two sites:
  - **Subgranular zone (SGZ) of the dentate gyrus** → excitatory granule cells, which stay local
  - **Subventricular zone (SVZ)** of the lateral ventricle → neuroblasts travel the rostral migratory stream (chain migration along astrocyte guides, §17) → inhibitory olfactory bulb interneurons (granule and periglomerular cells)
- Same principle as in the embryo: multipotent stem cells with astrocytic (glial) character (§15)
  - SVZ lineage: astrocytic stem cell → transit-amplifying cells (expansion) → neuroblasts, in a niche beside blood vessels
  - Some progeny die, some become glia, a substantial minority become granule cells indistinguishable from embryonic-born ones
- **Niche restriction**: precursors able to make neurons in culture exist in many adult regions (cortex, spinal cord), yet neurogenesis normally happens only in SGZ and SVZ → local permissive factors, not the absence of precursors, limit it (→ repair, §27)

Life course of an adult-born granule cell — each stage reruns a developmental step:

1. **Proliferation** (§15): progenitors in the SGZ divide

2. **Fate commitment** (§16): daughters commit to granule-cell identity (Prox-1, later NeuN), decided soon after birth
3. **~1 week old**: immature, few spines; GABA depolarizing; no glutamatergic input yet
4. **~2 weeks old**: spines and first glutamatergic inputs; low-threshold Ca^{2+} spikes and a *lower threshold for LTP* = highly plastic, easily recruited
5. **Maturation and integration**: dendrites and spines grow, axon reaches CA3, mature excitability → part of the cortico-hippocampal circuit

- **Survival window** (§19): many newborn cells die in their first weeks (peak death around 1–2 weeks) → activity in this window decides who stays
- Integration → new neurons compete with existing ones for synapses → activity-dependent rewiring (§22)
- Up-regulated by: running, enriched environment, hippocampus-dependent learning, LTP (§21), NMDA-receptor blockade
- Down-regulated by: stress and animal models of depression, aging, irradiation
- Blocking adult neurogenesis degrades certain hippocampus- and olfactory bulb-dependent behaviours (§22–23)

Tools used in the experiments below:

- **BrdU** = thymidine analogue built into the DNA of dividing cells → marks cells born at injection time
  - Count soon after injection → *proliferation*
  - Count weeks later → *survival*
  - Co-label with NeuN/Prox-1 → *neuronal differentiation*
- **X-ray irradiation** of the head → kills dividing precursors → ablates neurogenesis
- **Voluntary wheel running** → increases proliferation → enhances neurogenesis

21. LTP enhances neurogenesis

Core idea: Synaptic plasticity itself drives neurogenesis. Inducing LTP at perforant path → DG synapses increases the *proliferation* of progenitor cells over the following days and the *survival* of 1–2-week-old granule cells, without changing what they differentiate into.

- Two separate effects:
  - **Proliferation ↑**: more dividing progenitors in the days after LTP, larger at several days than at one day → a promitotic signal that builds up after induction rather than an immediate effect
    - Extra cells persist → more surviving new neurons weeks later
  - **Survival ↑**: cells born 1–2 weeks *before* LTP (the critical death window) survive better → a direct prosurvival effect
- **No effect on differentiation**: the same proportion become neurons → fate is fixed early (Notch/bHLH decision, §16) → LTP acts selectively, not instructively (§15)
- Effect requires LTP, not just stimulation: a pseudotetanus (same pulses, no LTP) changes nothing → cause = strong depolarization of LTP, not electrode or synaptic activation per se

Why 1-week-old cells survive without being potentiated — LTP acts via the surrounding tissue:

- 2-week-old cells: some express **Zif268** (LTP-induced immediate early gene, needed for late LTP) after LTP → they receive enough glutamatergic input to be potentiated directly
- 1-week-old cells: no Zif268 (no glutamatergic input yet), but survival still increases → effect must be indirect:
  1. LTP occurs in surrounding *mature* granule cells
  2. Potentiated cells and glia release factors — BDNF (up for at least a day after LTP), extrasynaptic glutamate, GABA
  3. These create a pro-survival microenvironment (BDNF is a neurotrophin: TrkB → PI3K–Akt brake on apoptosis, §19)
  4. Young neurons in their critical window survive instead of dying

- Proliferation: strong depolarization / glutamate excites precursor cells via NMDA-receptor-dependent L-type Ca^{2+} channel signalling (*excitation–neurogenesis coupling*), possibly with secreted mitotic signals
- Apparent paradox: *global* NMDA-receptor blockade increases neurogenesis, yet NMDA-dependent LTP also increases it → blocking all NMDA receptors ≠ local, pathway-restricted NMDA activation
- Implication: LTP during learning may be the signal for learning-induced neurogenesis

Evidence (design): LTP induced on one side only, so the other hemisphere is a within-animal control; BrdU given after LTP (counted next day → proliferation; weeks later → survival of those cells) or 1–2 weeks before LTP (counted weeks later → survival of cells already born).

22. Neurogenesis conserves memory capacity by clearing LTP

Core idea: After LTP saturation, hippocampal memory capacity returns only as the saturated LTP decays back to baseline — and that decay is driven by ongoing neurogenesis. Less neurogenesis → LTP persists → capacity recovers slowly; more neurogenesis → LTP decays faster → capacity recovers faster.

- Recovery of learning tracks LTP decay step by step:
  - LTP saturated → contextual fear learning impaired
  - LTP partly decayed → learning partly recovered
  - LTP back at baseline (about two weeks) → learning fully recovered
- Neurogenesis sets the speed of that decay:
  - Neurogenesis ablated (irradiation) → LTP still elevated when controls are back at baseline → learning still impaired
  - Neurogenesis enhanced (running) → LTP decays sooner → learning recovers sooner
- Across individual animals: the more residual potentiation, the less freezing (negative correlation) → residual LTP, not time itself, limits learning

- The same holds after whole-brain saturation by repeated electroconvulsive shocks (recovery delayed by irradiation)

How new neurons renew the circuit — the proposed mechanism:

1. Newborn granule cells integrate into the existing DG circuit (§20)
2. They **compete** with older neurons for synaptic inputs and outputs
3. Competition drives activity-dependent **rewiring** → older potentiated connections are replaced or weakened
4. Net potentiation slowly **renormalizes to baseline** = slow homeostatic downscaling (§14)
5. Old memory traces are **cleared from the hippocampus** — not lost: the memory becomes neocortex-dependent (systems consolidation)
6. New neurons contribute **naive, flexible synapses** that can be recruited into new assemblies → **capacity for new learning restored** ("keeps the gate open for new memories")

- Converse: blocking integration of new synapses **stabilizes** the hippocampal network → old memories stay hippocampus-dependent longer and DG-LTP persists longer, but less room for new learning
- Specificity: neurogenesis renormalizes strong, long-lasting (tetanus-like) LTP, i.e. the consolidated potentiation that sleep-mediated downscaling does not remove

Evidence (design): one hippocampus is lesioned so that learning depends on the other, making animals sensitive to its saturation; the perforant path → DG pathway of the remaining side is saturated with repeated tetani in awake rats, LTP is followed over days, and contextual fear conditioning is trained at different delays. Irradiation (fewer new neurons) or wheel running (more) is added; running does not help when neurogenesis is also ablated → running acts via neurogenesis. Auditory fear learning and general motility are unaffected throughout → the deficit is in hippocampal learning capacity.

23. Integrated picture: the plasticity–neurogenesis loop

Core idea: LTP and adult neurogenesis regulate each other in a loop. Learning-induced LTP produces more young neurons; those neurons help encode new, distinct memories and, as they integrate, erase old potentiation — so the hippocampus stays able to learn.

The loop — how the DG keeps encoding without filling up:

1. Learning → LTP at DG synapses (§9)
2. LTP → more progenitor proliferation + better survival of young neurons (§21)
3. Young (≈2-week-old) neurons have a low LTP threshold → readily recruited into new memory assemblies = **encoding role**
4. As they integrate, they compete with older neurons → circuit rewired → old LTP decays (§22) = **clearing role**
5. Old memories move out of the hippocampus (become cortex-dependent); synaptic weight returns to baseline
6. Capacity for the next learning event restored → back to step 1

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Each pass strengthens synapses for a new memory and later clears them, so overall synaptic weight stays below saturation.

- **Pattern separation**: adult-born neurons appear especially important
  - Stimulating neurogenesis improves discrimination of closely similar contexts
  - Silencing all granule cells *except* adult-born ones does not impair pattern separation → newborn cells carry it
  - Fits the computational view: neurogenesis increases capacity for new information while removing old information
- **Aging**: neurogenesis declines with age → slower clearance of old potentiation → hippocampus closer to saturation → less capacity to acquire new information → candidate cause of age-related memory decline
  - Exercise enhances neurogenesis and learning in aged animals; boosting neurogenesis is being explored against age-related memory loss

24. Axon injury: degeneration and its spread through the circuit

Core idea: Cutting an axon (*axotomy*) dooms its distal segment through an active degeneration program (Wallerian degeneration). It also stresses or kills the cell body, which loses its target-derived trophic supply, and harms the neuron's partners, so damage spreads through the circuit.

- **Distal segment**: synaptic transmission fails → after a delay, membrane breaks down and cytoskeleton is degraded = Wallerian degeneration
  - Active, regulated process, not passive withering
- **Cell body**:
  - May die by apoptosis → cut off from target-derived trophic factors (§19)
  - Otherwise *chromatolysis*: soma swells, nucleus moves off-center, rough ER fragments; protein and RNA synthesis rise and gene expression changes → reversible if regeneration succeeds
- **Debris clearance** differs, which matters for regrowth (§25):
  - PNS: Schwann cells break up and engulf myelin, divide, recruit blood macrophages, and secrete growth factors → fast
  - CNS: oligodendrocytes cannot clear myelin; the blood–brain barrier keeps macrophages out → only resident microglia → slow; myelin debris persists
- **Spread to partners**:
  - Targets that lose their main input atrophy, sometimes die
  - *Synaptic stripping*: terminals on the injured neuron withdraw and are replaced by glial processes (microglia, astrocytes) → less synaptic drive
  - Result: degeneration propagates anterogradely and retrogradely → injury at one site affects distant regions
- Reactive astrocytes form a *glial scar* near the injury (§25)

Evidence: in the Wlds (Wallerian degeneration slow) mutant mouse, distal stumps of cut nerves survive for weeks instead of days → degeneration is an actively regulated response that can be slowed.

25. Why central axons regenerate poorly

Core idea: Peripheral axons regrow to their targets; adult central axons sprout briefly and stall. The CNS environment lacks growth promoters and contains inhibitors (myelin proteins, the glial scar), and adult central neurons have a weakened intrinsic growth program. The poor regeneration of the mature CNS is the exception, not the rule.

Peripheral regeneration — why it works (reuses embryonic axon-growth mechanisms):

1. Distal axon degenerates, but its connective-tissue sheath and Schwann cells survive
2. Sprouts from the proximal stump enter the distal stump
3. Schwann cells secrete chemotropic and trophic factors; laminin in their basal lamina and Ig-superfamily adhesion molecules support growth; inhibitors in the perineural sheath keep axons on track = **a guided channel**
4. Axons reach targets, form new endings, are remyelinated; chromatolysis reverses

- Not perfect: some axons reach wrong targets → strength recovers better than fine movement

Central failure: short sprouts end in swollen *retraction bulbs*; long-distance growth is rare

- It is the adult CNS that is unusual: embryonic and immature mammalian CNS axons, and fish/frog CNS axons, regenerate

Barrier	Where it comes from	How it blocks growth	Intervention tried
Too few promoters	CNS has little laminin, low trophic factors	No support for elongation	Neurotrophin infusion, laminin conduits, Schwann cell grafts → slightly more growth, little function
Blocked cytokine signal	SOCS3 in retinal ganglion neurons	Suppresses CNTF–gp130 growth signal	SOCS3 deletion → more optic nerve regeneration

Barrier	Where it comes from	How it blocks growth	Intervention tried
Myelin inhibitors	Central myelin debris	Nogo, MAG, OMgp → shared receptors NogoR (with p75) and PirB	PirB deletion → modestly more corticospinal regrowth
Glial scar	Reactive astrocytes	Physical barrier + CSPGs → PTPσ → Rho/ROCK	Chondroitinase; early anti-inflammatory drugs (prednisolone)
Weak intrinsic program	Adult central neurons	cAMP and GAP-43 fall postnatally	Raise cAMP or GAP-43

- Shared receptors for several myelin inhibitors → blocking one pathway could neutralize many inhibitors at once
- The scar is also adaptive: limits injury size, restores the blood–brain barrier, reduces inflammation
- Myelin inhibition fits development: myelination happens after axon extension is complete

Conditioning lesion — shows the intrinsic program can be reawakened:

1. A DRG neuron has a peripheral branch (regrows well) and a central branch (regrows poorly)
2. The peripheral branch is cut first = **primer**
3. This switches on the growth program (cAMP up, GAP-43)
4. When the central branch is cut days later, it regrows past the lesion

- Mimicked by raising cAMP or GAP-43 alone

Why the adult CNS gave this up ("Faustian bargain")

- The mammalian brain remodels its wiring during postnatal critical periods, then must stabilize it so brief perturbations don't rewire it
- The stabilizers (myelin at the end of critical periods, astrocyte proteoglycans) also block regrowth → poor regeneration is the price of precise, stable circuits

Evidence: peripheral nerve segments grafted into the CNS let central axons grow centimeters, while optic nerve segments grafted into a peripheral nerve stop peripheral axons → the environment decides much of the outcome. Central axons still grow less than peripheral ones along the same graft → an intrinsic limit as well.

26. Recovery without regeneration: rewiring intact circuits

Core idea: Some function returns even when cut axons never regrow, because intact axons sprout and form new connections around the lesion. This latent plasticity is strongest in early critical periods but can be reawakened by injury in the adult.

Intraspinal detour — how cortex regains access to motor neurons below a spinal lesion:

1. Corticospinal axons are cut → cortex can no longer reach motor neurons below the lesion
2. Over weeks, intact corticospinal axons above the lesion sprout new terminal branches
3. Sprouts synapse onto propriospinal interneurons whose axons run around the lesion = **the bypass**
4. These interneurons contact motor neurons below the lesion → partial recovery of function

- Similar reorganization occurs in motor cortex and brain stem
- Some benefits of grafts may come from such rewiring rather than from regrowth of cut axons
- Most promising: growth-promoting interventions combined with behavioural therapy that steers rewiring
- Risk: promoting growth can also create maladaptive circuits

27. Replacing lost cells

Core idea: In most of the CNS the neurons we are born with are the only ones we get, so lost neurons must be replaced from outside or coaxed from precursors. Adult neurogenesis (§20) is the proof of principle: new neurons can differentiate, grow processes through adult neuropil and integrate into working circuits.

- Two lessons taken from adult neurogenesis:
  - Endogenous new neurons integrate → transplanted neurons or precursors might too
  - Precursors can be induced to divide and differentiate → their output might be boosted to replace lost neurons

Transplantation

- Embryonic neurons (e.g. dopaminergic grafts into the putamen in Parkinson disease): promising, but hard to obtain in number and often too mature to integrate
- Neural precursors (stem cells, committed precursors): often become neurons typical of the *transplant site*, not their origin
  - → the local environment instructs neuronal type (instructive signals, §15)
  - But precursor plasticity is limited → differentiation must be started in culture before grafting
- **Embryonic stem (ES) cells**: pluripotent, can be expanded in large numbers and directed to specific fates with developmental signals (e.g. spinal motor neurons for ALS, dopaminergic neurons for Parkinson disease)
- **Induced pluripotent stem (iPS) cells**: patient skin fibroblasts reprogrammed to pluripotency
  - No embryos needed; immunologically matched to the patient
  - Directed to motor neurons with retinoic acid and hedgehog signals; already used for drug screens

Stimulating endogenous neurogenesis

- Precursors exist outside the niches (cortex, spinal cord) but normally make no neurons there (§20)

- Trauma or ischemia stimulates neurogenesis even in cortex and spinal cord, but too little to repair; growth factors can boost it in animals

Replacing glia

- Oligodendrocyte loss → myelin stripping continues after injury → axons that were not cut also lose function
- Endogenous remyelination is insufficient (e.g. multiple sclerosis)
- ES-derived oligodendrocyte precursors grafted into injured spinal cord → remyelination and better locomotion in animals

The end point is functional synapses

- Regenerated or new neurons are useful only if they form working synapses with targets
- Combined approaches are likely needed (e.g. chondroitinase + neurotrophic factors; cellular bridges + drugs blocking inhibitors where axons leave the bridge)

Evidence: a peripheral nerve grafted from the eye to the superior colliculus lets retinal axons regrow into it; flashing light on the eye then evokes action potentials in collicular neurons → regenerated adult axons can form functional synapses.